Higher moment theory and learnability of bosonic states

Paper

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Slides

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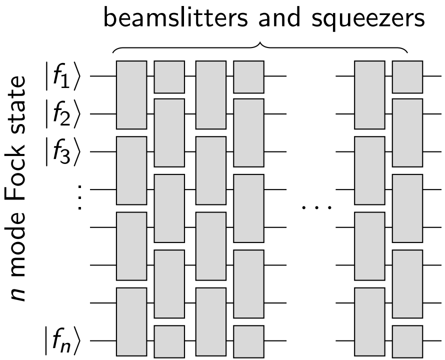
Joint work with Y. Wang, A. Gorshkov, C. Oh, B. Fefferman, I. Datta, S. Ghosh

Introduction

- Learning (properties of) quantum states as efficiently as possible is a fundamental task in quantum information
- There is a lot of recent work on learning continuous variable (CV) states (eg. Mele, Mele, et al. 2024; Zhao, Liao, et al. 2024; Bittel, Mele, et al. 2025; Fanizza, Iyer, et al. 2025)
- Most *sample and time efficient* learning algorithms work for **Gaussian** or **near-Gaussian** states
- Learning a generic n-mode CV state requires $\sim$ (energy)^n samples (Mele, Mele, et al. 2024)

Our goal

- Our goal is to find interesting, **highly non-Gaussian** classes of CV states with *sample and time efficient* learning algorithms
- Motivated by an open problem proposed in Aaronson and Grewal 2023, we start with **BosonSampling states**



Background

- A bosonic system of n modes is described by the creation and annihilation operators $a_1^\dagger, \ldots, a_n^\dagger, a_1, \ldots, a_n$
- Equivalently, by position and momentum operators $\boldsymbol{r} = (x_1, \ldots, x_n, p_1, \ldots, p_n)$

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- A unitary transformation is called *Gaussian* if it acts affinely on $\boldsymbol{r}$

$$\mathcal{U}_{S,\boldsymbol{d}}^\dagger r_i \mathcal{U}_{S,\boldsymbol{d}} = \sum_j S_{ij} r_j + d_i$$

- **The group of Gaussian unitaries is** $\mathrm{Sp}(2n, \mathbb{R}) \ltimes \mathbb{R}^{2n}$

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- We will ignore the $\mathbb{R}^{2n}$ part as it is easily dealt with

Problem statement (Aaronson and Grewal 2023)

- Let $|\boldsymbol{f}\rangle$ denote an unknown Fock state on n modes, $\boldsymbol{f} = (f_1, \ldots, f_n) \in \mathbb{N}_0^n$
- Let $\mathcal{U}_S$ denote a Gaussian unitary specified by an unknown matrix $S \in \mathrm{Sp}(2n, \mathbb{R})$
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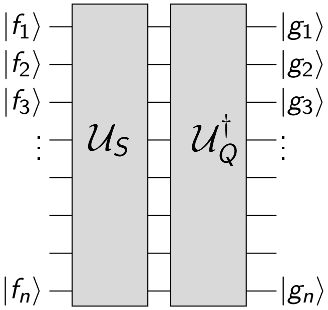
- Given N copies of ψ, we want to learn ψ to ε precision
- Find a $\boldsymbol{g}$ and Q such that $|\langle \boldsymbol{g}| \mathcal{U}_Q^\dagger \mathcal{U}_S |\boldsymbol{f}\rangle| \geq 1 - \varepsilon$



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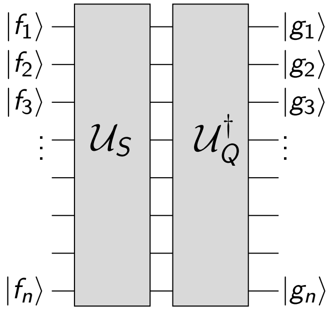

- How large must N be? *Want it to be* $\leq \mathrm{poly}(\text{energy})$
- What is the measurement scheme? *Want it to be practically accessible*
- How hard is the classical postprocessing? *Want it to be* $\leq \mathrm{poly}(n)$

Analogous fermionic problem

- The analogous fermionic problem was solved in Aaronson and Grewal 2023
- Crucial difference: fermionic Fock states are Gaussian, bosonic Fock states are highly non-Gaussian

First main result

- We will solve this problem with *moments* for $|\psi\rangle = \mathcal{U}_S |\boldsymbol{f}\rangle$

$$\Lambda^{(t)}_{i_1,\ldots,i_t;j_1,\ldots,j_t} = \langle\psi| r_{i_1} \ldots r_{i_t} r_{j_1} \ldots r_{j_t} |\psi\rangle$$

- Suppose we know $\Lambda^{(1)\prime}, \Lambda^{(2)\prime}$ such that $\|\Lambda^{(t)\prime} - \Lambda^{(t)}\| \leq \varepsilon_t$

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Theorem

We find an **efficient** *algorithm that finds a Q and $\boldsymbol{g}$ such that*

$$|\langle\boldsymbol{f}|\mathcal{U}_S^\dagger\mathcal{U}_Q|\boldsymbol{g}\rangle| \geq 1 - \mathcal{O}\left(\varepsilon_1 n^2 \|\boldsymbol{f}\|_\infty^3 \|S^T S\|^{3/2} + \varepsilon_2 n^2 \|\boldsymbol{f}\|_\infty \|S^T S\|^3\right)$$

This can therefore be combined with *any* strategy to measure moments to yield a **sample and time efficient learning algorithm**

Measuring moments

Theorem

Using a **median of means** *estimator,* $\Lambda^{(t)}$ *for* $\mathcal{U}_S\,|\boldsymbol{f}\rangle$ *can be estimated to norm precision* ε *with probability at least* $1-\delta$ *using* $N \sim \mathcal{O}\left(\frac{\log(n/\delta)}{\varepsilon^2}\left(n\|f\|_\infty\|S^TS\|\right)^{2t}\right)$ **heterodyne** *samples.*

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Corollary (Full solution to Aaronson and Grewal 2023)

N *heterodyne samples from the state* $\mathcal{U}_S |\boldsymbol{f}\rangle$ *suffice via our* **efficient** *algorithm to learn a* $Q, \boldsymbol{g}$ *satisfying* $|\langle \boldsymbol{f}| \mathcal{U}_S^\dagger \mathcal{U}_Q |\boldsymbol{g}\rangle| \geq 1-\gamma$ *with probability at least* $1-\delta$*, where*

$$N \sim \mathcal{O}\left(\frac{\log(n/\delta)}{\gamma^2} n^8 \|\boldsymbol{f}\|_\infty^{10} \|S^T S\|^{10}\right)$$

Roughly, $n\,\|\boldsymbol{f}\|_\infty\,\|S^T S\| \sim$ energy

Overview

G_t states

Definition

A G_t state is a thermal (or ground) state of a (non-degenerate) Hamiltonian that is degree t in the quadrature operators

- A G_t state is **fully specified** by its first t moments

Pure G_t states

- Suppose ψ is a G_t state; it is the ground state of $H = \sum_i \alpha_i \hat{M}_i$ where $\hat{M}_i$ are degree $\leq t$ operators

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- Because H is non-degenerate, any state with the matching moments is ψ
- If H is *gapped*, then **inverse poly moment precision corresponds to inverse poly state infidelity** (Yu and Wei 2023)

Relevance to the learning algorithm

- A Fock state $|\boldsymbol{f}\rangle$ is a G_4 state; it is the ground state of $H = \sum_i (\hat{a}_i^\dagger \hat{a}_i - f_i)^2$
- The set of G_t states is closed under the action of Gaussian unitaries (because they act affinely)

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- Therefore, $\mathcal{U}_S |\boldsymbol{f}\rangle$ is the ground state of a gapped degree 4 Hamiltonian
- Knowing the **first four moments to inverse poly precision** suffices to learn the state
- This guarantees *sample efficiency*

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- Therefore, $\mathcal{U}_S |\boldsymbol{f}\rangle$ is the ground state of a gapped degree 4 Hamiltonian
- Knowing the **first four moments to inverse poly precision** suffices to learn the state
- This guarantees *sample efficiency*, but this does **not** guarantee a *time-efficient* learning algorithm

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Sketch of our algorithm

- For simplicity, we will restrict our attention here to

  passive Gaussian unitaries (specified by $W \in \mathrm{U}(n)$)

  the Fock state $|1, \ldots, 1\rangle \equiv |1^n\rangle$

  the error free case (assume moments are known *perfectly*)

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- We sketch the learning algorithm for $\mathcal{U}_W |1^n\rangle$ with these simplicifcations

Passive moment transformations

- Recall $W \in U(n)$ acts as

$$\mathcal{U}_W^\dagger a_i \mathcal{U}_W = \sum_j W_{ij} a_j$$

- How do moments **transform**?

$$\sigma^{(t)}_{i_1,\ldots,i_t;j_1,\ldots,j_t} = \langle 1^n | \mathcal{U}_W^\dagger a_{i_1} \ldots a_{i_t} a^\dagger_{j_1} \ldots a^\dagger_{j_t} \mathcal{U}_W | 1^n \rangle$$
$$(\sigma^{(t)}_0)_{i_1,\ldots,i_t;j_1,\ldots,j_t} = \langle 1^n | a_{i_1} \ldots a_{i_t} a^\dagger_{j_1} \ldots a^\dagger_{j_t} | 1^n \rangle$$
$$\sigma^{(t)} = W^{\otimes t} \sigma^{(t)}_0 W^{\dagger \otimes t}$$

- $\sigma^{(t)}$ is a submatrix of a unitary transformation of the $\Lambda^{(t)}$ moments

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- So now we use **fourth moments**, $\sigma^{(2)}$

Learning algorithm

Given the fourth moments $\sigma^{(2)} = W^{\otimes 2}\sigma_0^{(2)} W^{\dagger\otimes 2}$ for the state $\mathcal{U}_W \left|1^n\right\rangle$, determine W

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- Because $\sigma_0^{(2)}$ is a matrix on $(\mathbb{C}^n)^{\otimes 2}$, we can represent it in bra-ket notation using the standard basis $|1\rangle, \ldots, |n\rangle$ of $\mathbb{C}^n$

$$\sigma_0^{(2)} = 4(\mathbb{I} + U_{\mathrm{SWAP}}) - 2\sum_{i=1}^{n} |i,i\rangle\langle i,i| ,$$

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- Denote the i^{th} column vector of W by $|w_i\rangle$. Given access to $\sigma^{(2)}$, we can thus form the matrix

$$\begin{aligned} A &= \frac{1}{2}\left(4(\mathbb{I} + U_{\mathrm{SWAP}}) - \sigma^{(2)}\right) \\ &= \sum_{i=1}^{n} (|w_i\rangle \otimes |w_i\rangle)(\langle w_i| \otimes \langle w_i|), \end{aligned}$$

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Given access to a matrix $A = \sum_{i=1}^{n}(|w_i\rangle \otimes |w_i\rangle)(\langle w_i| \otimes \langle w_i|)$, learn $|w_i\rangle$ (up to phases and permutations)

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- Unitarily diagonalize A and take the $+1$ eigenvalues — we learn the eigenspace $\operatorname{span}\{|w_i\rangle \otimes |w_i\rangle \mid i = 1, \ldots, n\}$
- A numerical diagonalization routine will return the vectors

$$|\tilde{w}_i\rangle = \sum_{j=1}^{n} U_{ij} |w_j\rangle \otimes |w_j\rangle = \sum_{j=1}^{n} |U_{ij}| e^{\mathrm{i}\phi_{ij}} |w_j\rangle \otimes |w_j\rangle$$

- Take the intersection of the Schmidt decompositions of all $|\tilde{w}_i\rangle$ to learn all the $|w_i\rangle$ up to phases

Learning algorithm

- We have therefore learned the matrix $V = W\Phi P$, where Φ is some arbitrary diagonal unitary matrix, and P is some arbitrary permutation matrix
- It follows that the state $\mathcal{U}_V \left|1^n\right\rangle$ is the same as $\mathcal{U}_W \left|1^n\right\rangle$ up to an irrelevant global phase, thereby completing the learning algorithm

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- To prove the main result, we need to track how the error in the moments propagates through the algorithm to an error in the fidelity

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- Solves open problem left by Aaronson and Grewal 2023
- We construct an **infinite set of Gaussian invariants for generic bosonic states** using classical invariant theory of the action of symplectic group on moments, relevant for state conversion (Chabaud, Markham, et al. 2020; Hahn, Ferrini, et al. 2024)
- Lots of possible extensions!

Extensions

- Our algorithm generalizes to a slightly larger class of states, and fails interestingly in other places
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  - Our algorithm requires $N \sim \mathrm{poly}(n, \|\boldsymbol{f}\|_\infty, \|S^T S\|)$ samples
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Paper

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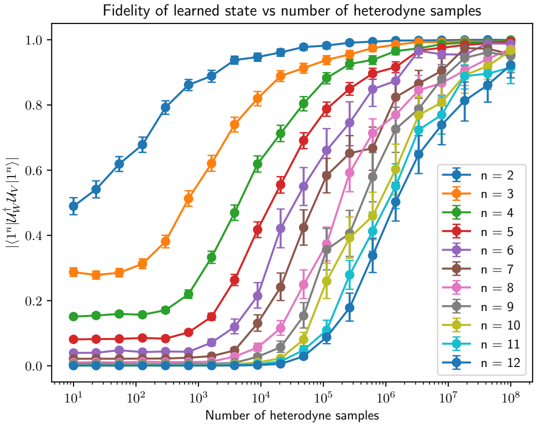
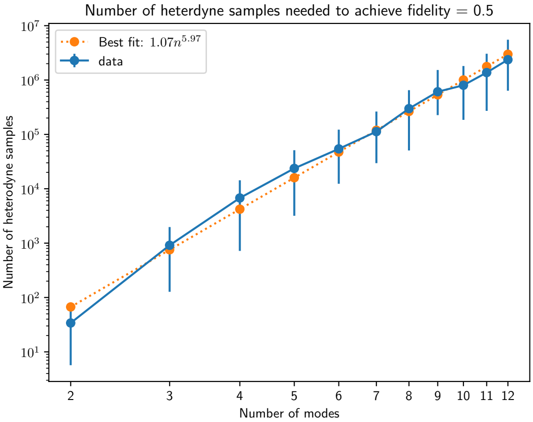
Additional slides

Numerics



Fidelity of learned state vs number of heterodyne samples
$|\langle 1^n|\mathcal{U}_W^\dagger \mathcal{U}_V|1^n\rangle|$
Number of heterodyne samples
n = 2
n = 3
n = 4
n = 5
n = 6
n = 7
n = 8
n = 9
n = 10
n = 11
n = 12



Number of heterdyne samples needed to achieve fidelity = 0.5
Best fit: $1.07n^{5.97}$
data
Number of heterodyne samples
Number of modes

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